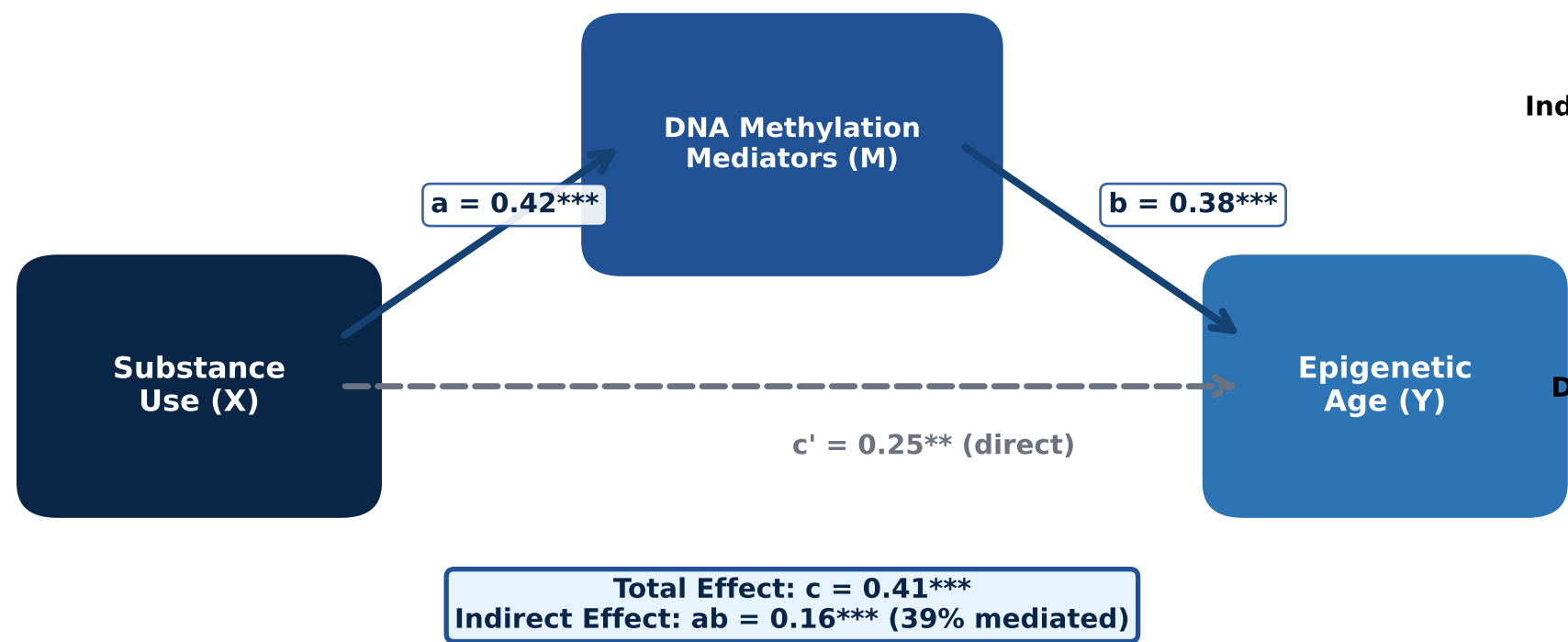
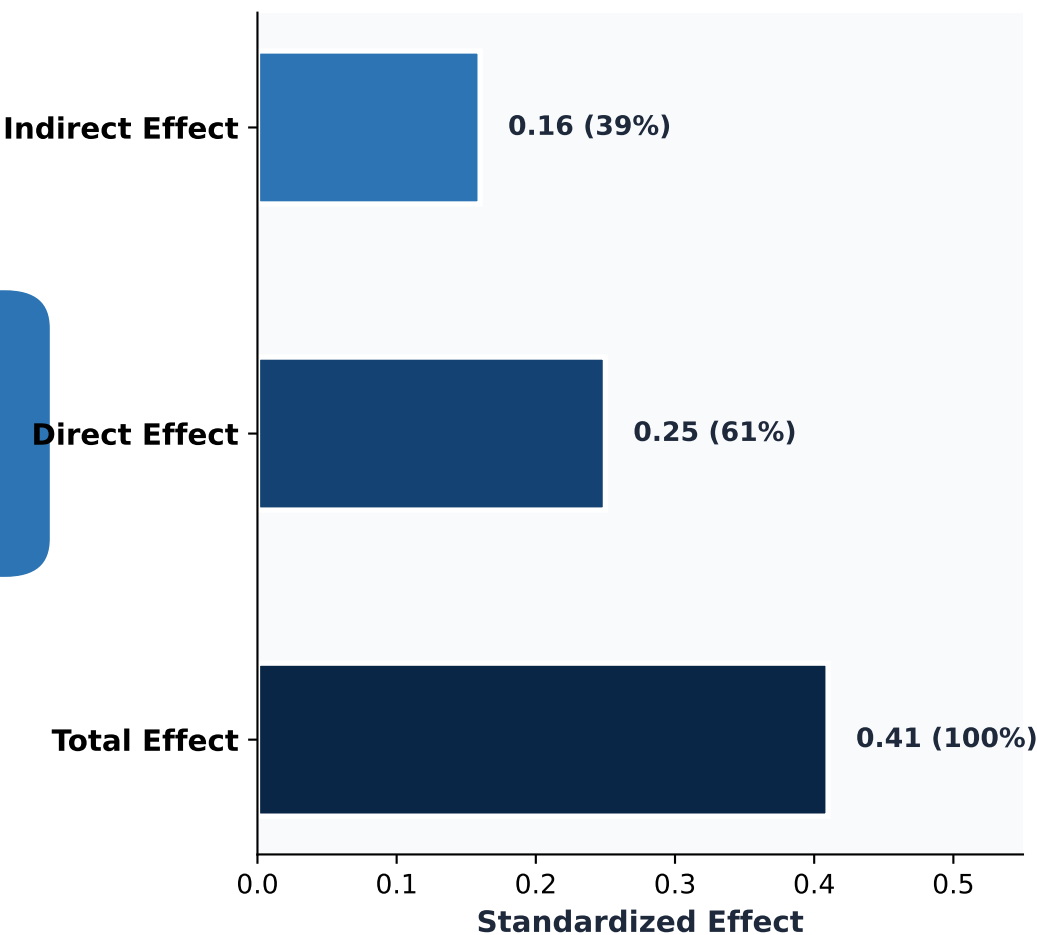


Figure 3. Mediation Analysis: Substance Use, DNA Methylation, and Epigenetic Aging

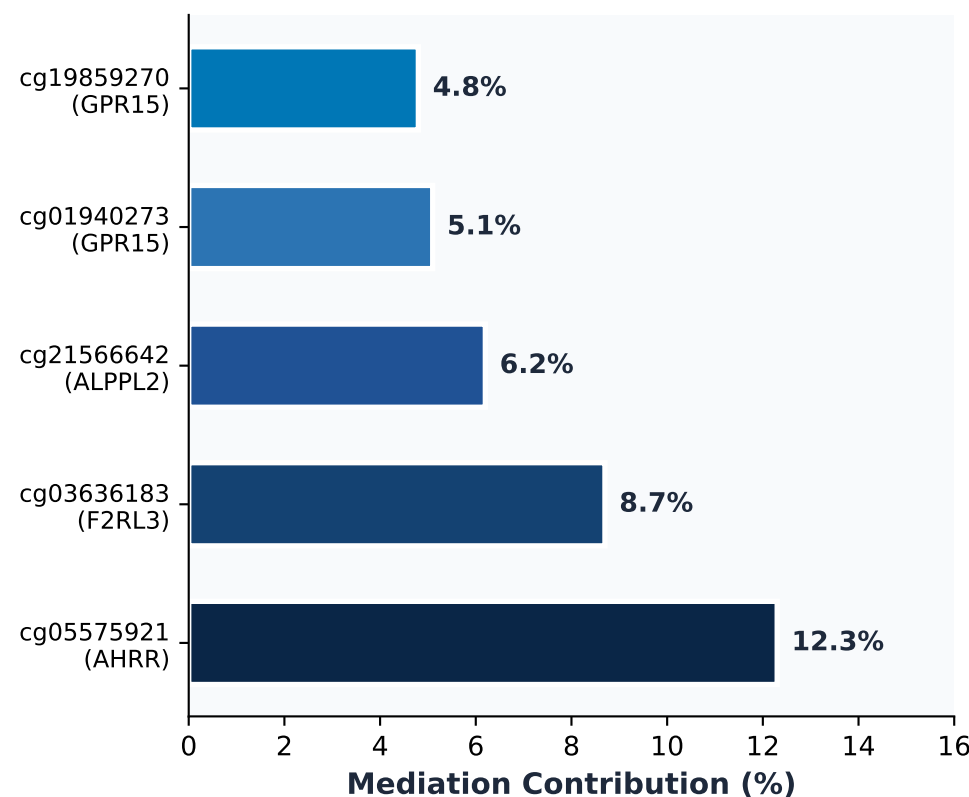
A. Mediation Pathway Model



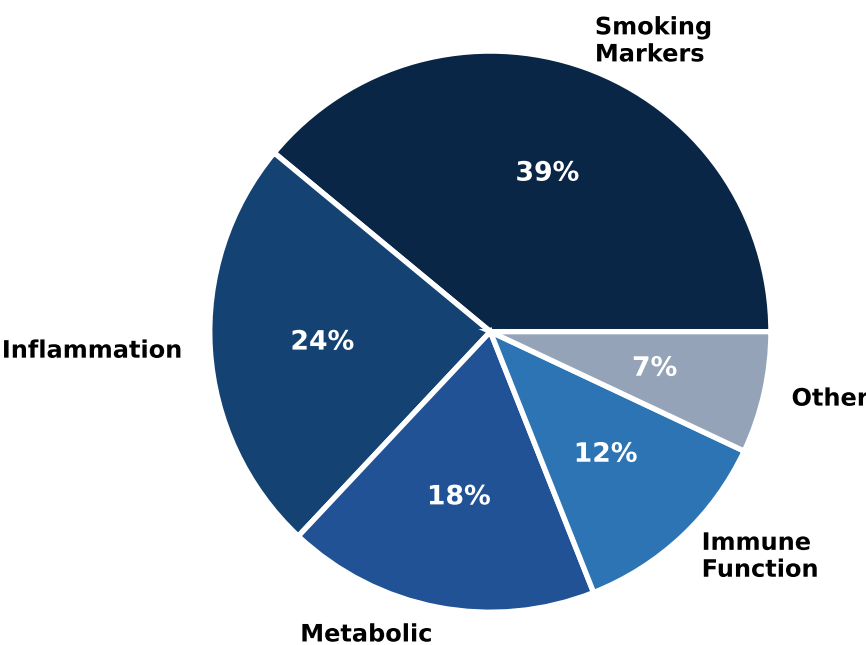
B. Effect Decomposition



C. Top Mediating CpG Sites



D. Biomarker Contributions



E. Bootstrap Validation

MEDIATION ANALYSIS RESULTS	
Bootstrap: 10,000 iterations Confidence: 95% Bias-corrected	
Indirect Effect (ab): Point estimate: 0.160 95% CI: [0.118, 0.205] p < 0.001 ***	
Proportion Mediated: 39.0% 95% CI: [28.8%, 50.0%]	
Sobel Test: z = 8.42, p < 0.001	
*** p < 0.001, ** p < 0.01, * p < 0.05	